

# CRYSTALdataGen Manual

Jurij

May 19, 2026

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# Chapter 1

## Project Overview

### 1.1 Scientific Goal

The goal of the CRYSTALdataGen project is to build a workflow for generating machine-learning interatomic potentials capable of reproducing:

- Total energies
- Atomic forces
- Vibrational frequencies
- Born effective charges (BECs)
- Infrared (IR) intensities
- Phonon spectra

The workflow combines:

- CRYSTAL reference calculations
- Distorted structure generation
- MACE/MACELES model training
- Analytical Hessian evaluation
- IR spectrum reconstruction
- Mode matching against reference phonons

### 1.2 Repository Workflow

The project workflow consists of the following stages:

1. Reference structure preparation
2. Random distortion generation
3. Single-point DFT calculations
4. HDF5 dataset construction
5. XYZ export for MACE training

6. Hyperparameter sweep training
7. Model evaluation
8. Phonon and IR analysis
9. Mode matching with CRYSTAL references

## Chapter 2

# Input Data and Reference Calculations

### 2.1 Reference Structures

The workflow starts from optimized crystal structures stored as CIF files.

Reference structures may originate from:

- CRYSTAL geometry optimization
- Experimental CIFs
- Standardized structures

The code automatically handles:

- Primitive cells
- Supercells
- Symmetry standardization
- CIF normalization

### 2.2 CRYSTAL Reference Calculations

CRYSTAL is used to compute:

- Total energies
- Atomic forces
- Hessians
- Born effective charges
- Infrared intensities

Important CRYSTAL outputs:

- `freq.out`
- `*.hessfreq`
- `*.born`

## 2.3 Reference Database

Reference data are stored inside:

`ref_db.h5`

Stored quantities include:

- Frequencies
- Intensities
- Hessians
- Born effective charges
- Dielectric tensors

## Chapter 3

# Random Distortion Generation

### 3.1 Purpose of Distortions

Distorted structures are generated to sample the local potential energy surface around equilibrium.

The generated structures are used for:

- Force training
- Hessian learning
- Vibrational robustness
- Data augmentation

### 3.2 Random Cartesian Displacements

Atomic positions are displaced randomly:

$$\mathbf{r}'_i = \mathbf{r}_i + \Delta \mathbf{r}_i \quad (3.1)$$

Each Cartesian displacement component is sampled from:

$$\Delta r_{i\alpha} \sim \mathcal{U}(-d_{\max}, d_{\max}) \quad (3.2)$$

where:

- $d_{\max}$  is the maximum displacement amplitude
- $\alpha \in \{x, y, z\}$

### 3.3 Random Strain Distortions

A random symmetric strain tensor is generated:

$$\varepsilon_{\alpha\beta} = \frac{1}{2} (A_{\alpha\beta} + A_{\beta\alpha}) \quad (3.3)$$

with:

$$A_{\alpha\beta} \sim \mathcal{U}(-s_{\max}, s_{\max}) \quad (3.4)$$

The strained lattice becomes:

$$\mathbf{h}' = \mathbf{h}(\mathbf{I} + \boldsymbol{\varepsilon}) \quad (3.5)$$



### 3.4 Mixed Distortions

Mixed distortions combine:

- Random atomic displacements
- Random strain

$$\mathbf{r}'_i = \mathbf{r}_i + \Delta\mathbf{r}_i \quad (3.6)$$

$$\mathbf{h}' = \mathbf{h}(\mathbf{I} + \boldsymbol{\varepsilon}) \quad (3.7)$$

### 3.5 Short-Distance Rejection

Distortions are rejected if atoms become too close:

$$d_{ij} < d_{\min}^{(ij)} \quad (3.8)$$

This prevents:

- Unrealistic forces
- Divergent SCF cycles
- Numerical instability

## Chapter 4

# Training Dataset Construction

### 4.1 HDF5 Dataset Layout

Training data are stored as:

```
/structures/<name>/reference  
/structures/<name>/distortions/<id>
```

Each distortion stores:

- Energy
- Forces
- Stress
- Atomic positions
- Cell vectors

### 4.2 Unit Conventions

Important unit conversions:

| Quantity     | Unit              |
|--------------|-------------------|
| Energy       | eV                |
| Forces       | eV/Å              |
| Hessian      | eV/Å <sup>2</sup> |
| Frequency    | cm <sup>-1</sup>  |
| IR intensity | KM/mol            |

### 4.3 XYZ Export

The HDF5 database is converted into:

- train.xyz
- validation.xyz

These files are consumed directly by MACE.

## Chapter 5

# MACELES Model Training

### 5.1 Model Architecture

The workflow uses:

- MACE
- MACELES

MACELES extends MACE with:

- Latent Ewald summation
- Long-range electrostatics
- Born effective charge prediction

### 5.2 Training Loss

The total training loss is:

$$\mathcal{L} = w_E \mathcal{L}_E + w_F \mathcal{L}_F + w_\sigma \mathcal{L}_\sigma \quad (5.1)$$

where:

- $\mathcal{L}_E$  is the energy loss
- $\mathcal{L}_F$  is the force loss
- $\mathcal{L}_\sigma$  is the stress loss

### 5.3 Hyperparameter Sweeps

The workflow automatically sweeps:

- Cutoff radius  $r_{\max}$
- Energy weights
- Force weights
- Batch size
- Random seeds

## Chapter 6

# Phonons and Vibrational Analysis

### 6.1 Analytical Hessian

The Hessian matrix is:

$$H_{i\alpha,j\beta} = \frac{\partial^2 E}{\partial r_{i\alpha} \partial r_{j\beta}} \quad (6.1)$$

### 6.2 Mass Weighting

The Hessian is transformed into the dynamical matrix:

$$D_{i\alpha,j\beta} = \frac{H_{i\alpha,j\beta}}{\sqrt{m_i m_j}} \quad (6.2)$$

### 6.3 Diagonalization

The vibrational modes satisfy:

$$\mathbf{D}\mathbf{q}_\lambda = \omega_\lambda^2 \mathbf{q}_\lambda \quad (6.3)$$

where:

- $\omega_\lambda$  are vibrational frequencies
- $\mathbf{q}_\lambda$  are mass-weighted eigenvectors

### 6.4 Mass Unweighting

Cartesian normal modes are obtained using:

$$\mathbf{e}_{i\alpha,\lambda} = \frac{\mathbf{q}_{i\alpha,\lambda}}{\sqrt{m_i}} \quad (6.4)$$

The modes are normalized:

$$\mathbf{e}_\lambda \leftarrow \frac{\mathbf{e}_\lambda}{\|\mathbf{e}_\lambda\|} \quad (6.5)$$

## 6.5 Frequency Conversion

Frequencies are converted into  $\text{cm}^{-1}$ :

$$\tilde{\nu}_\lambda = \frac{\omega_\lambda}{2\pi c} \frac{1}{100} \quad (6.6)$$

Imaginary modes correspond to:

$$\omega_\lambda^2 < 0 \quad (6.7)$$

## Chapter 7

# Born Effective Charges and IR Intensities

### 7.1 Born Effective Charges

The Born effective charge tensor is:

$$Z_{i,\alpha\beta}^* = \frac{\partial P_\alpha}{\partial u_{i\beta}} \quad (7.1)$$

where:

- $P_\alpha$  is the polarization
- $u_{i\beta}$  is the atomic displacement

### 7.2 Acoustic Sum Rule

The BEC tensors are corrected using:

$$\tilde{Z}_i^* = Z_i^* - \frac{1}{N} \sum_j Z_j^* \quad (7.2)$$

### 7.3 Projection onto Normal Modes

Mode effective charges are computed as:

$$\mathbf{Z}_\lambda = \sum_i \mathbf{Z}_i^* \mathbf{e}_{i\lambda} \quad (7.3)$$

### 7.4 IR Intensities

Infrared intensities are proportional to:

$$I_\lambda \propto |\mathbf{Z}_\lambda|^2 \quad (7.4)$$

## 7.5 Spectrum Broadening

The discrete spectrum is broadened using Gaussians:

$$G(\nu) = \sum_{\lambda} I_{\lambda} \exp \left[ -\frac{(\nu - \nu_{\lambda})^2}{2\sigma^2} \right] \quad (7.5)$$

with:

$$\sigma = \frac{\text{FWHM}}{2\sqrt{2\ln 2}} \quad (7.6)$$

# Chapter 8

## Mode Matching

### 8.1 Purpose

Mode matching compares ML phonons against CRYSTAL reference phonons.

### 8.2 Overlap Matrix

The overlap matrix is:

$$O_{\lambda\mu} = \left| \mathbf{e}_{\lambda}^{\text{ref}} \cdot \mathbf{e}_{\mu}^{\text{ML}} \right| \quad (8.1)$$

### 8.3 Hungarian Assignment

Matching minimizes the cost matrix:

$$C_{\lambda\mu} = 1 - O_{\lambda\mu} \quad (8.2)$$

### 8.4 Degenerate Subspace Matching

Degenerate subspaces are compared using:

$$S(A, B) = \frac{\|\mathbf{U}_A^T \mathbf{U}_B\|_F}{\sqrt{\min(n_A, n_B)}} \quad (8.3)$$

### 8.5 Diagnostics

The workflow evaluates:

- Frequency error
- Mode overlap
- Intensity similarity
- Imaginary mode count



## Chapter 9

# Evaluation Workflow

### 9.1 Evaluation Pipeline

Each trained model is evaluated by:

1. Geometry optimization
2. Hessian evaluation
3. Frequency calculation
4. BEC extraction
5. IR intensity reconstruction
6. Mode matching
7. Spectrum comparison

### 9.2 Composite Score

The composite score combines:

- Frequency MAE
- Weighted frequency MAE
- Spectrum L2 error
- Pearson correlation
- Imaginary mode penalty

# Chapter 10

## Code Organization

### 10.1 Main Scripts

| Script                    | Purpose               |
|---------------------------|-----------------------|
| run_master.py             | Sweep orchestration   |
| inference.py              | Evaluation pipeline   |
| distortions.py            | Structure distortions |
| build_train_db.py         | Dataset generation    |
| build_ref_db.py           | Reference database    |
| export_train_valid_xyz.py | XYZ export            |

### 10.2 Recommended Improvements

Recommended future improvements:

- Add workflow diagrams
- Add automatic unit validation
- Add schema versioning
- Add structured logging
- Add CI tests for parsers
- Add automatic dataset statistics
- Add database provenance tracking
- Add reproducibility manifests

## Chapter 11

# Future Work

Potential future extensions:

- Raman intensities
- Finite-temperature phonons
- Anharmonic corrections
- Active learning
- Multi-functional datasets
- Cross-material transfer learning